Tomas Monopoli

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Location: Aalborg, Denmark

Website | LinkedIn | GitHub | Google Scholar | Web of Science

Core Competences: *Physics-Informed AI, Bayesian Inference & Statistics, Uncertainty Quantification (UQ), Gaussian Process Regression (GPR/Kriging), Deep Learning (Transformers, CNNs), Power Electronics, Digital Twins, Predictive Maintenance, EMC and Space Systems.*

Technical Skills: *Coding* | Python (PyTorch, TensorFlow, Scikit-learn, numpy, pandas), MATLAB, LaTeX, C++ | *Simulation & Tools* | Ansys HFSS, COMSOL, SPICE, Git, Docker, Linux, VS Code

Career Summary [List of Publications]

PhD Engineer with 6+ years of experience in physical modeling and AI-driven system optimization. Bridging the gap between theoretical physics and industrial applications through robust machine learning pipelines and advanced statistical methods. Proven track record in leading high-impact research projects for energy and space sectors.

Experience

Postdoc in AI and Power Electronics | AAU, Aalborg, DK — 2025 – Now
- Leading research for the PHY-CALIPER Project to discover "unknown physics" and calibrate predictive models for next-gen energy systems.
- Optimizing predictive maintenance workflows using **statistical ML**, **PIML**, and **Bayesian methods**.

Visiting Research Fellow | European Space Agency (ESA) / ESTEC, Noordwijk, NL — 2021
- Developed data-driven **wideband EM modeling** for CubeSats, replacing heavy full-wave simulations with efficient equivalent models to **accelerate EMC testing** (ESA project).

R&D Engineer | Edison, Milan, IT — 2019 – 2020
- Orchestrated R&D innovation projects, bridging technical feasibility with corporate business strategy.

Summer Internship | CERN (CESP Program), Geneva, CH — 2019

Education

PhD in Electrical Engineering (cum Laude) | Politecnico di Milano, IT — 2020 – 2025
- **Joint Research with ESA/ESTEC**: Developed AI-driven optimization frameworks for modeling satellite electromagnetic environments, replacing physical testing with data-driven predictions.

Alta Scuola Politecnica (ASP) | PoliMi & PoliTo, Milan/Turin, IT — 2018 – 2020
- **Excellence Program**: Selected among the top 1% of students for a double-degree program focused on multidisciplinary innovation.

MSc in Electrical Engineering (110/110 cum Laude) | PoliMi & PoliTo — 2017 – 2020
BSc in Electrical Engineering (110/110) | PoliMi — 2014 – 2017

Technical Projects

AI-powered OCR for Scientific Papers | Project Link | GitHub — 2024
- Developed a pipeline using **Transformer architectures** to accurately extract complex **LaTex** expressions and structured data from PDFs. Optimized for **RAG (Retrieval-Augmented Generation)** workflows with LLMs.

SmartBin: Waste Classification System | Entrepreneurship Program — 2018 – 2019
- Built a computer-vision-equipped bin using **CNNs** on Raspberry Pi; The prototype was awarded a presentation slot at **Italian Tech Week** in Turin.

Scientific Engagement

- **Technical Reviewer** for scientific journals (TEMC, LECPA, IEEE Access) and AEES 2023 conference
- **Presenter** at conferences (ESA workshop on Aerospace EMC 2025, COMPO 2024, GEMCCON 2020) and **Session Chair** for PTSET 2023